AGRIculture

farming + community

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What is AGRIculture?

A social design for urban gardening in the Troy food desert and to create a community around such knowledges.

Inspirations

- Growing up in the NYC food desert and learning to garden in an urban environment.
- Grandparents' neighborhood, where each family grows one crop (squash, grapes, beans, tomatoes, etc.) and they all trade.
    - Sense of community and exchange of food, seeds, and knowledge.

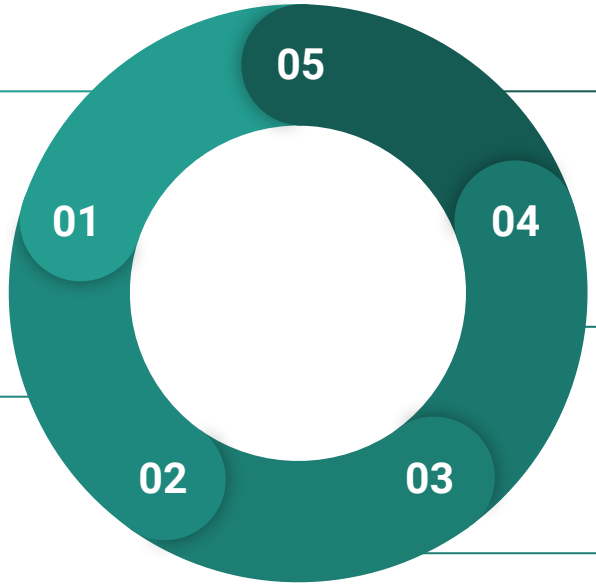
Food Deserts

- As defined by the USDA, a “food desert” is recognized as a part of the country vapid of fresh fruit, vegetables, and other healthful whole foods, and is usually an impoverished area.
    - Less than a third of residents in these locales do not own cars and live in excess of a mile from a store which provides fresh whole foods.
- Systematic issue due to distribution and bad industry practices.
    - Also climate change.
- I can’t really solve this huge issue over one summer. But it can be alleviated.

The GADIE Design Process

Goals
Education of agricultural practices, community building, and improved produce access in food deserts.

Analysis
There are other organizations and other hydroponics system already in the community. Stakeholders want accessible education. Ethnography.



Evaluation
Is the knowledge able to be spread without me? How well can the design be reproduced? Degree of documentation?

Implementation
Partnership with the NATURE Lab. Resources on the Sanctuary website or radio.

Design
Iterations of the hydroponics system and the accompanying materials for education.

Iterations

- **1:** Instructions for a cheap, modular system made available on the Internet.
  - Achieves one goal of information accessibility, but doesn't solve the whole problem.
  - Problems: There are already dozens of DIY hydroponics systems out there already. What's missing is the education and community around it.
- **2:** Partnering with The Sanctuary for Independent Media.
  - Has access to resources and social network has been established.
  - Problems: They're wary of another foray into hydroponics. Differing needs between stakeholders.
  - Takeaway: Maybe there isn't a need to design a totally new system, but a need for engagement.

Proposed Solutions

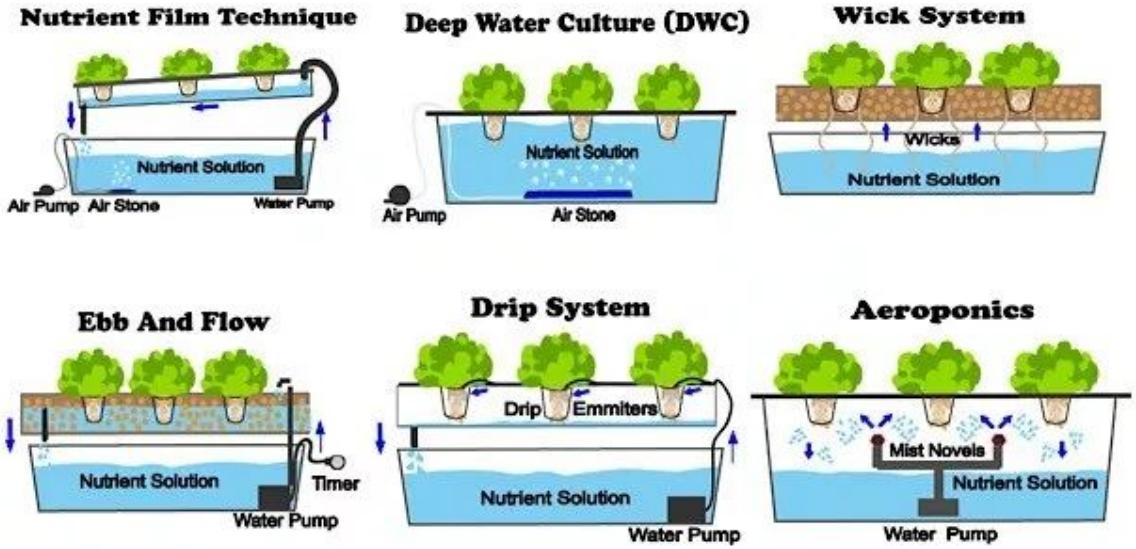
After iterating and community feedback...

Accessible gardening through hydroponics.

Community building through workshops.

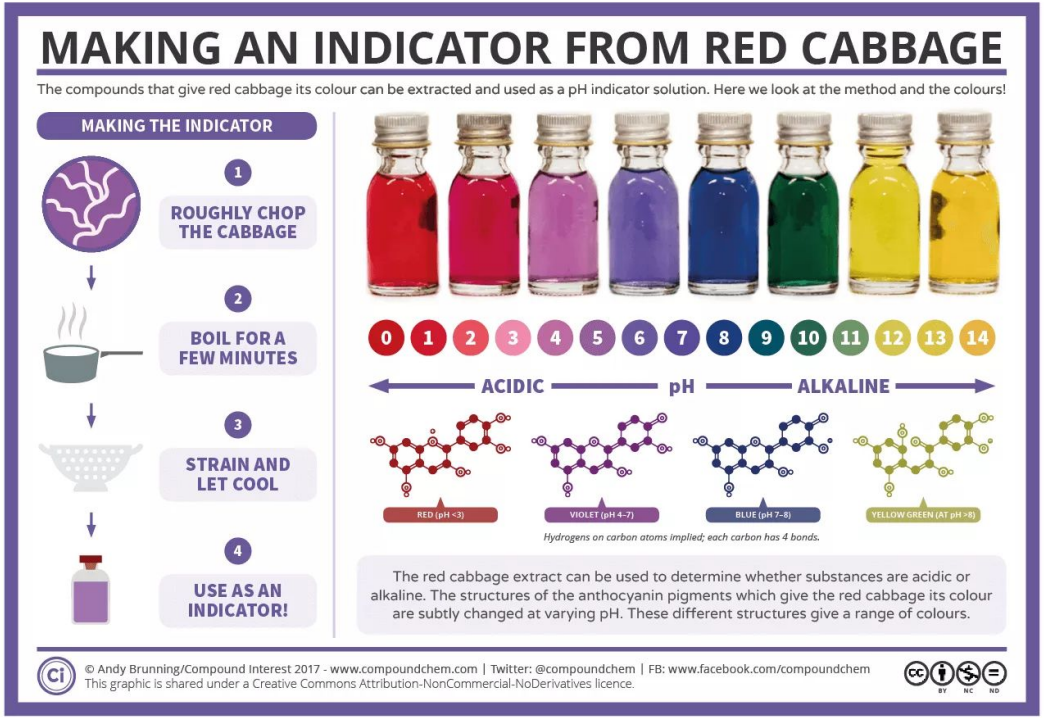
Workshop #1: Components of a Hydroponic System

- Using the Sanctuary's large system, run a demo.
- Types of hydroponic systems: wick, water culture, ebb and flow.
- Choosing substrate.



Workshop #2: pH testing

- Making pH testing strips using red cabbage.
- Understanding the role of pH in crop growth.
- Ideal pH is slightly acidic (5.8-6.2).
    - Should turn purple.



Workshop #3: Choosing Crops

- Learn about different subsistence crops and their needs.
  - Nutritional value?
  - Is it something that your family wants to eat?
- Tomatoes, lettuce, spinach, microgreens, peppers, etc.
  - Which hydroponics systems work best for each of these?
  - Summer crops vs. winter crops.
- Maybe with the Kitchen Sanctuary and its classes on making nutritional food?

Workshop #4: Building your own system

- Skills and instructions on constructing a family-scale system.
  - Different designs for different crop needs, etc.
- Building off skills from previous workshops and DIY projects/resources available online.
- Using items that are cheap and accessible.
  - Stuff from hardware stores, Walmart, etc.
- Option for using the large NATURE Lab system.

Workshop #5: Potluck

- Food produced by the NATURE Lab as well as family gardens.
  - At the end of the program, get to share fruit (or vegetable) of your labors.
- Opportunity for people to get to know each other and share knowledge.
  - Inspired by the community and number of people who showed up to the Spirit of the Suffragettes.

Thank you.
Questions?